## PHASE EXTERNAL DESCRIPTION:

The scheduler manages all tasks in the system using a dynamic linked list of Task Control Blocks, or TCB. Each TCB stores important information such as the name, task name, execution state, and pointer to the next task in the list. When the create_task() function creates a new task, TCB allocates and initializes the task with a unique ID and READY. The scheduler maintains a head pointer and a current pointer.

The Yield() function manages task execution, using round-robin. The scheduler advances the current point to the next task when the task yields, going back to the head when the end of the list is reached. The scheduler skips tasks not in the READY state until it is READY. Once ready, it is marked as RUNNING. Tasks are terminated with kill_task(), which means the task state is DEAD. The garbage collector, garbage_collect(), removes the DEAD tasks from the linked list and deallocates the memory.

A binary semaphore is used to provide mutual exclusion for shared resources, with each semaphore given a resource name. 1 or 0, and a queue is used. 1 means the resources are available. 0 means the resources are in use. The down() operation requests access to the resource, and if the value of the semaphore is 1, then it is set to 0, and the task can run. If the semaphore is 0, the task is placed into the queue, and the state is BLOCKED. The up() operation releases the resource, and if there are tasks in the queue, the next task is moved into the READY state.

Both have the dump() function to display, for the scheduler, the current state, ID, and name of all tasks, and, for the semaphore, the current resource name, value, and waiting queue.